Pimpri Chinchwad Education Trust's

Pimpri Chinchwad College of Engineering

Department of Computer Engineering

SY B. Tech.

LAB MANUAL

Advanced Data Structures Laboratory

BCE24PC02

Prepared By

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	Pimpri Chinchwad Education Trust's **Pimpri Chinchwad College of Engineering**
Department of Computer Engineering	

Institute Vision:

To be one of the top 100 Engineering Institutes of India in coming five years by offering exemplarily Ethical, Sustainable and Value Added Quality Education through a matching ecosystem for building successful careers.

Institute Mission:

M1. Serving the needs of the society at large through establishment of a state-of-art Engineering Institute

M2. Imparting right Attitude, Skills, Knowledge for self-sustenance through Quality Education

M3. Creating globally competent and Sensible engineers, researchers and entrepreneurs with an ability to think and act independently in demanding situations

Department Vision

To be a premier Computer Engineering department by achieving excellence in Academics and Research for creating globally competent and ethical professionals.

Department Mission

M1: To develop technologically competent and self-sustained professionals through contemporary curriculum.

M2: To nurture innovative thinking and collaborative research, making a positive impact on society.

M3: To provide a state-of-the art computing environment and learning opportunities through the Center of Excellence.

M4: To foster leadership skills and ethics with holistic development.

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Program Educational Objectives (PEO'S)

PEO NO	Program Educational Objectives Statements
PEO1	Graduates of the Programme will leverage Engineering fundamentals in their successful careers
PEO2	Graduates of the Programme will exhibit innovation and technological competence in computing all through their careers.
PEO3	Graduates of the Programme will perform to the expectations of employers, explore the opportunities for higher studies and/or show the awareness of self-business challenges.
PEO4	Graduates of the Programme will demonstrate ethical practices, lifelong learning with the awareness of needs of the society.

Program Outcomes (POs) (from 2024-25 as per new NBA guidelines) Engineering Graduates will be able to

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one’s own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program specific Outcomes:

PSO 1) Graduates will be able to apply problem solving skills through programming techniques for addressing real life problems using suitable Data Structures and Algorithms.

PSO 2) Graduates will be able to apply Software Development Life Cycle and knowledge of Data Analytics, Artificial Intelligence, Computer Vision, Web Technology, Computer Network & Security to provide innovative computing solutions.

Each experiment must include:

1	Problem Statement
2	Objectives:
3	Course Outcome Mapping:
4	Theory
5	Algorithm
6	Program
7	Sample Output
8	Conclusion:

Assignment No. 1

Aim:

Consider the Dictionary Implementations which allow for efficient storage and retrieval of key-value pairs using binary search trees. Each node in the tree store (key, value) pair. The dictionary should support the following operations efficiently:

1. Insert word (Handle insertion of duplicate entry)
2. Delete word
3. Search specific word
4. Display dictionary (Traversal)
5. Mirror image of dictionary
6. Create a copy of dictionary
7. Display dictionary level wise

Objectives:

- To understand the concept of **Binary Search Trees (BST)**
- To implement **dictionary operations using BST**
- To analyze efficiency of operations like search, insert, and delete
- To apply tree traversal techniques
- To understand deep copy and mirror operations

Course Outcome Mapping:

Theory: ***Students are expected to explore and write***

Algorithm:

Insert Operation

1. If tree is empty → create new node
2. Compare key with root
3. If key < root → insert in left subtree
4. If key > root → insert in right subtree
5. If key == root → update value (handle duplicate)

Delete Operation

1. Search the node
2. Case 1: Leaf node → delete directly

3. Case 2: One child → replace node with child
4. Case 3: Two children → replace with inorder successor

Search Operation

1. Start from root
2. Compare key
3. Traverse left or right accordingly
4. If found → return value

Traversal (Inorder)

- Left → Root → Right
- Displays dictionary in sorted order

Mirror Image

- Swap left and right child of every node recursively

Copy Dictionary

- Perform deep copy using recursion

Level-wise Display

- Use Queue (BFS traversal)

Program:

Sample Output:

Conclusion:

Assignment No. 2

Aim:

Implement a multiplayer game system that uses an AVL tree data structure to organize and manage player data efficiently. The multiplayer game supports multiple players participating simultaneously, and the AVL tree is used to store player information such as player_id and scores (key, value pair). The system should provide following operation:

1. Player Registration
2. Leaderboard Display
3. Remove player from game

Objectives:

- To understand AVL Trees (self-balancing BSTs)
- To implement player data management using AVL Tree
- To maintain a balanced tree for efficient operations
- To simulate a leaderboard system
- To perform insertion, deletion, and traversal operations

Course Outcome Mapping:

Theory: ***Students are expected to explore and write***

Algorithm:

1. Insert node as BST.
2. Update height.
3. Calculate balance factor.
4. Perform rotations if unbalanced:
    - LL Rotation
    - RR Rotation
    - LR Rotation
    - RL Rotation

Program:

Sample Output:

Conclusion:

Assignment No. 3

Aim:

Consider the scenario of a maze. The maze is represented as a grid of cells, where each cell can be either open or blocked. Each cell in the maze represents a vertex. The goal is to find a path from the starting point to the goal within a given maze using DFS and BFS.

Objectives:

- To understand **graph traversal algorithms (DFS & BFS)**
- To represent a maze as a **graph (grid-based structure)**
- To compare DFS and BFS approaches
- To find a valid path from **start to goal node**

Course Outcome Mapping:

Theory: ***Students are expected to explore and write***

Algorithm:

DFS Algorithm

1. Start from source cell
2. Mark it as visited
3. Explore all unvisited neighbors recursively
4. If goal is found → stop
5. Backtrack if needed

BFS Algorithm

1. Insert start cell into queue
2. Mark as visited
3. While queue is not empty:
    - Remove front element
    - Check if goal reached
    - Add all unvisited neighbors to queue
4. Track path using parent mapping

Program:

Sample Output:

Conclusion:

Assignment No. 4

Aim:

Represent a graph of your college campus using adjacency list /adjacency matrix. Nodes should represent the various departments and links should represent the distance between them. Find a minimum spanning tree using Kruskal's algorithm or using Prim's algorithm.

Objectives:

- To understand graph representation techniques
- To model real-world problems (campus map) using graphs
- To implement Minimum Spanning Tree algorithms
- To optimize total distance between departments

Course Outcome Mapping:

Theory: ***Students are expected to explore and write***

Algorithm:

Prim's Algorithm

1. Start from any vertex
2. Select minimum weight edge connecting visited and unvisited node
3. Repeat until all vertices are included

Kruskal's Algorithm

1. Sort all edges in increasing order
2. Pick smallest edge
3. Check for cycle using Union-Find
4. Add edge if no cycle
5. Repeat until MST formed

Program:

Sample Output:

Conclusion:

Assignment No. 5

Aim:

Consider the scenario of the supply chain management system. The supply chain involves multiple locations, such as suppliers, warehouses, distribution centers, and retail stores, connected by transportation networks. Find the shortest possible route using Dijkstra's algorithm to optimize transportation path and inventory management.

Objectives:

- To understand graph-based modeling of supply chains
- To implement Dijkstra's shortest path algorithm
- To optimize transportation routes
- To minimize cost/time in logistics systems

Course Outcome Mapping:

Theory: ***Students are expected to explore and write***

Algorithm (Dijkstra's):

1. Initialize distance array with infinity
2. Set source distance = 0
3. Mark all nodes as unvisited
4. Select node with smallest distance
5. Update distances of its neighbors
6. Repeat until all nodes are processed

Program:

Sample Output:

Conclusion:

Assignment No. 6

Aim:

Implement the scenario of a file system which maintains directory structure using the Red-Black Tree. Each node in the tree represents a directory, and the tree is balanced to ensure efficient insertion, deletion, and display operations when navigating through the file system.

Objectives:

- To understand Red-Black Trees (self-balancing BSTs)
- To simulate a file system directory structure
- To perform efficient insert, delete, and display operations
- To maintain tree balance automatically

Course Outcome Mapping:

Theory: ***Students are expected to explore and write***

Algorithm :

Insert Directory

1. Insert node like BST (as Red)
2. Fix violations:
    - Case 1: Recoloring
    - Case 2: Rotation
3. Ensure root remains Black

Delete Directory

1. Perform BST deletion
2. Fix tree using rotations and recoloring

Display Directory Structure

- Use Inorder Traversal (sorted view)
- Can also use Level Order for hierarchy

Program:

Sample Output:

Conclusion: